# Session 7: Nested Loops & Pattern Printing

C Language Study Notes & Code Guide

## Core Concepts: Nested Loops

A **nested loop** is a loop inside another loop.

- The **outer loop** controls the rows.
- The **inner loop** controls the columns (elements printed per row).
- After each row is complete, a newline character (`\n`) is printed to move to the next line.

## 1 Task 1: 5x5 Instagram Post Feed Grid

**Notes:** The outer loop runs 5 times (rows), and the inner loop prints 5 symbols per row (columns).

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 5; i++) {         // rows
        for (int j = 1; j <= 5; j++) {     // columns
            printf(" ");
        }
        printf("\n");                      // new line after each row
    }
    return 0;
}
```

**Sample Output:**

## 2 Task 2: Right-Angled Triangle (Leaderboard Rank Numbers)

**Notes:** Row `i` contains exactly `i` numbers. The inner loop condition depends on the outer loop variable (`j <= i`). Numbers increase continuously across rows.

```c
#include <stdio.h>

int main() {
    int num = 1;   // continuous counter (like rank numbers)

    for (int i = 1; i <= 5; i++) {         // rows
        for (int j = 1; j <= i; j++) {     // i numbers in row i
```

```
            printf("%d ", num);
            num++;                          // next rank
        }
        printf("\n");
    }
    return 0;
}
```

**Sample Output:**

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

## 3 Task 3: Star Pyramid (6 Rows, Centered)

**Notes:** A centered pyramid requires three parts per row: leading spaces + stars + newline.

- **Spaces:** `rows - i` (pushes stars toward the center).
- **Stars:** `2 * i - 1` (odd numbers: 1, 3, 5, 7... keep the pyramid symmetric).

```c
#include <stdio.h>

int main() {
    int rows = 6;

    for (int i = 1; i <= rows; i++) {
        // Print leading spaces for centering
        for (int j = 1; j <= rows - i; j++) {
            printf(" ");
        }
        // Print stars (odd count: 1, 3, 5, ...)
        for (int k = 1; k <= 2 * i - 1; k++) {
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

## 4 Task 4: Alternating 0s and 1s Grid (4x4 Checkered)

**Notes:** The value depends on row + column parity: if `(i + j) % 2 == 0`, print `0`, otherwise print `1`. This creates a checkerboard effect.

```c
#include <stdio.h>

int main() {
    int rows = 4, cols = 4;

    for (int i = 1; i <= rows; i++) {          // rows
        for (int j = 1; j <= cols; j++) {      // columns
            if ((i + j) % 2 == 0)
                printf("0 ");
            else
                printf("1 ");
        }
        printf("\n");
    }
    return 0;
}
```

**Sample Output:**

```
0 1 0 1
1 0 1 0
0 1 0 1
1 0 1 0
```

## 5 Task 5: Pyramid with User-Input Rows

**Notes:** Combines the pyramid logic from Task 3 with user input via `scanf`. Includes basic input validation to reject values less than 1.

```c
#include <stdio.h>

int main() {
    int rows;

    printf("Enter the number of rows for the pyramid: ");
    scanf("%d", &rows);

    if (rows < 1) {
        printf("Please enter a number greater than 0.\n");
        return 1;
    }

    for (int i = 1; i <= rows; i++) {
        // Leading spaces for centering
        for (int j = 1; j <= rows - i; j++) {
            printf(" ");
        }
        // Stars: 1, 3, 5, ... (2*i - 1)
        for (int k = 1; k <= 2 * i - 1; k++) {
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

## 6 Quick Revision Summary

| Task | Pattern Type | Key Logic |
|---|---|---|
| 1 | Full Grid | Inner loop fixed (`j <= 5`) |
| 2 | Triangle of Numbers | Inner loop depends on row (`j <= i`) |
| 3 | Centered Pyramid | Spaces = `n - i`, Stars = `2*i - 1` |
| 4 | Checkerboard | `(i + j) % 2` decides 0 or 1 |
| 5 | User-driven Pyramid | Same as Task 3 + `scanf()` validation |

**Golden Rule of Pattern Printing:** Outer loop controls rows, inner loop controls what gets printed in that row (spaces, symbols, or numbers), followed by exactly one `\n` at the end of each row.